

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2019-2020)

PRE-MEDICAL : ENTHUSIAST COURSE [PHASE-I]

TEST # 02

DATE : 21-04-2019

Test Type : Minor

Test Pattern : NEET

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **720**.
5. There are **three** parts in the question paper A,B,C consisting of **Physics, Chemistry** and **Biology** having **45 questions** each in Physics & Chemistry and **90 questions** in Biology part of equal weightage. Each question is allotted 4 (four) marks for **correct** response.
6. **One Fourth** mark will be deducted for indicated incorrect response of each question. **No deduction** from the total score will be made if no response is indicated for an item in the Answer Sheet.
7. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
8. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
9. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
10. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
11. **Do not fold or make any stray marks on the Answer Sheet.**

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MINOR TEST # 02

Topics

DATE : 21 - 04 - 2019

PRE-MEDICAL ENTHUSIAST COURSE : PHASE - I – PATTERN : [NEET]

PHYSICS : Geometrical Optics, Electrostatics, [Till Coulomb's Laws], NLM

CHEMISTRY : Organic Chemistry, Atomic Structure, Redox Reaction

BIOLOGY : Genetics [Till Cytoplasmic Inheritance], Cell Structure & Cell Cycle,
Human Reproduction [Till Gametogenesis], Endocrine System

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HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

1. The distance x covered in time t by a body having initial velocity v_0 and having a constant acceleration a is given by

$x = v_0 t + \left(\frac{1}{2}\right) at^2$. This result follows from :-

- (1) Newton's first law
(2) Newton's second law
(3) Newton's third law
(4) None of these

2. A material body A of mass m_1 exerts a force on another material body B of mass m_2 . If the acceleration of B be a_2 , the magnitude of the acceleration of A is :-

- (1) Zero (2) $m_2 a_2 / m_1$
(3) $m_1 a_2 / m_2$ (4) a_2

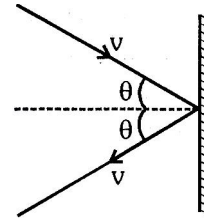
3. The velocity acquired by a mass m in travelling a certain distance d starting from rest under the action of a constant force is directly proportional to :-

- (1) $\sqrt{m}$ (2) m°
(3) $\frac{1}{\sqrt{m}}$ (4) m

4. Weight is defined as :-

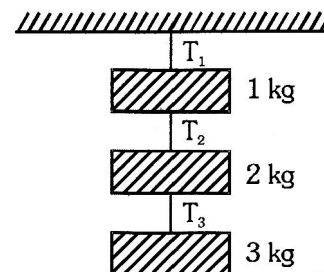
- (1) Force of attraction exerted by the earth
(2) Mass of a body
(3) Nature of a body
(4) None of these

5. A water jet, whose cross sectional area is 'a' strikes a wall making an angle ' θ ' with the normal and rebounds elastically. The velocity of water of density 'd' is v . Force exerted on wall is :-



- (1) $2av^2 d \cos \theta$
(2) $2av^2 d \sin \theta$
(3) $2avd \cos \theta$
(4) $avd \cos \theta$

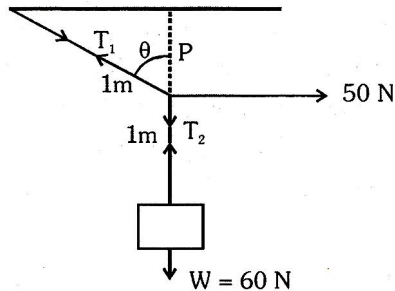
6. Find the tension T_2 for the system shown in fig.



- (1) $1g$ N
(2) $2g$ N
(3) $5g$ N
(4) $6g$ N

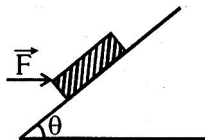
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7. A mass of 6 kg is suspended by a rope of length 2 m from a ceiling. A force of 50 N is applied in horizontal direction at the mid point of the rope. What is the angle between the rope and the vertical in equilibrium :-



- (1) $\tan^{-1}\left(\frac{4}{5}\right)$ (2) $\tan^{-1}\left(\frac{5}{4}\right)$
(3) $\tan^{-1}\left(\frac{5}{6}\right)$ (4) None

8. The figure shows a horizontal force $\vec{F}$ acting on a block of mass M on an inclined plane (angle θ). What is the normal reaction on the block ?

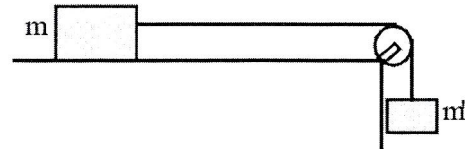


- (1) $mg\sin\theta + F\cos\theta$ (2) $mg\sin\theta - F\cos\theta$
(3) $mg\cos\theta - F\sin\theta$ (4) $mg\cos\theta + F\sin\theta$

9. The ratio of weights of a man in a stationary lift and in a lift accelerating downwards with a uniform acceleration is 3 : 2. The acceleration of the lift is:-

- (1) $\frac{g}{3}$ (2) $\frac{g}{2}$ (3) g (4) $\frac{4}{3}g$

10. In the fig. given below masses m and m' are tied with a thread passing over a pulley, m is on a frictionless horizontal surface. If acceleration due to gravity is g , the acceleration of m' in this arrangement will be :-



- (1) g (2) $m'g/(m + m')$
(3) mg/m' (4) $mg/(m - m')$

11. Two metallic spheres of radii 2 cm and 3 cm are given charges of -1×10^{-2} C and 6×10^{-2} C respectively. If these are connected by a conducting wire, the final charge on the bigger sphere is :-

- (1) 4×10^{-2} C (2) 1×10^{-2} C
(3) 2×10^{-2} C (4) 3×10^{-2} C

12. Charges of $10 \mu\text{C}$ each are placed at the corners of an equilateral triangle of side 5 cm. Then the force on each charge is :-

- (1) 78 N (2) 6.25 N
(3) 623.5 N (4) 22.5 N

13. A charge q is placed in the middle of two equal and like point charges Q . For this system to remain in equilibrium the value of q is :-

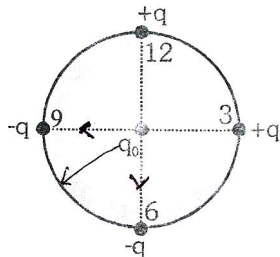
- (1) $-\frac{Q}{3}$ (2) $-\frac{Q}{4}$
(3) $\frac{Q}{2}$ (4) $-\frac{Q}{2}$

SPACE FOR ROUGH WORK

14. Two point charges placed at a distance V in air exert a force ' F '. The distance at which they exert same force when placed in a certain medium (dielectric constant K) is :-

(1) rK (2) r/K
(4) $r/\sqrt{K}$ (4) $r\sqrt{K}$

15. Four charges are placed at the circumference of the dial of a clock as shown in figure. If the clock has only hour hand, then the resultant force on a positive charge q_0 placed at the centre, points in the direction which shows the time as :-



(1) 1 : 30 (2) 7 : 30
(3) 4 : 30 (4) 10 : 30

16. A point object is placed at a distance of 10 cm and its real image is formed at a distance of 20 cm from a concave mirror. If the object is moved by 0.1 cm towards the mirror, the image will shift by about.

(1) 0.4 cm away from the mirror
(2) 0.4 cm towards the mirror
(3) 0.8 cm away from the mirror
(4) 0.8 cm towards the mirror

17. A ray of light passes through an equilateral glass prism in such a manner that the angle of incidence is equal to the angle of emergence and each of these angles is equal to $3/4$ of the angle of the prism. The angle of deviation is

(1) 45° (2) 39°
(3) 20° (4) 30°

18. The focal length of a convex lens is 10 cm and its refractive index is 1.5. If the radius of curvature of one surface is 7.5 cm, the radius of curvature of the second surface will be :-

(1) 7.5 cm (2) 15.0 cm
(3) 75 cm (4) 5.0 cm

19. The focal lengths of the objective and eye piece of a telescope are respectively 100 cm and 2 cm. The moon subtends an angle of 0.5° at the eye. If it is looked through the telescope, the angle subtended by the moon's image will be.

(1) 100° (2) 50°
(3) 25° (4) 10°

20. A man cannot see clearly the objects beyond a distance of 50 cm from his eyes. To see distant objects clearly he must use which kind of lenses and of what focal length

(1) 10 cm convex
(2) 100 cm concave
(3) 20 cm convex
(4) 50 cm concave

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21. A bird is at a height X above the water surface and a fish is at a depth Y below the surface. If the two are in a vertical line and μ is the refractive index of water w.r.t air, then distance of the fish as observed by the bird is

(1) $X + \frac{Y}{\mu}$ (2) $\mu X + \frac{Y}{\mu}$
(3) $\mu X + Y$ (4) $X + \mu Y$

22. A plano convex lens ($\mu = 1.5$) has radius of curvature 20 cm. It is silvered on its plane surface. Find focal length after silvering:-

(1) 10 cm (2) 20 cm
(3) 15 cm (4) 25 cm

23. A bubble in glass slab ($\mu = 1.5$) when viewed from one side appears at 6 cm and 4 cm from other side, then thickness of slab is:-

(1) 3.75 cm (2) 3 cm
(3) 15 cm (4) 2.5 cm

24. A ray of light travelling in air has wavelength λ , frequency n , velocity v and intensity I . If this ray enters into water then these parameters are λ' , n' , v' and I' respectively. Which relation is correct from following-

(1) $\lambda = \lambda'$ (2) $n = n'$
(3) $v = v'$ (4) $I = I'$

25. The magnifying power of a simple microscope is 6. The focal length of its lens in metres will be, if least distance of distinct vision is 25 cm

(1) 0.05 (2) 0.06 (3) 0.25 (4) 0.12

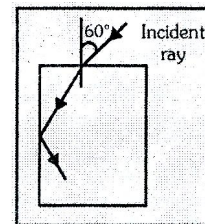
26. If the maximum acceptance angle of an optical fibre without cladding is $\theta = \sin^{-1}(\sqrt{3})$, then the refractive index of the optical fiber will be

(1) 2 (2) 1.5 (3) 1.33 (4) 3

27. A point source of light is placed 10 m below the surface of water of refractive index $5/3$. The minimum diameter of a disc which should be placed over the source on the surface of water to cut-off all light coming out of water is

(1) 2m (2) 6 m (3) 4 m (4) 15 m

28. For the given incident ray as shown in figure, the condition of total internal refraction of this ray the minimum refractive index of prism will be :



(1) $\frac{\sqrt{3}+1}{2}$ (2) $\frac{\sqrt{2}+1}{2}$

(3) $\sqrt{\frac{3}{2}}$ (4) $\frac{\sqrt{7}}{2}$

29. Two plane mirrors are at 45° to each other. If an object is placed between them then the number of images will be

(1) 5 (2) 9 (3) 7 (4) 8

SPACE FOR ROUGH WORK

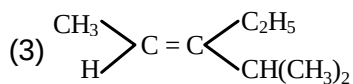
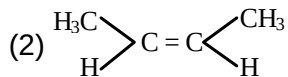
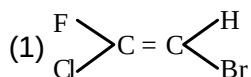
30. A microscope is focussed on a mark on a piece of paper and then a slab of glass of thickness 6 cm and refractive index 1.5 is placed over the mark. How should the microscope be moved to get the mark in focus again :-
 (1) 2 cm upward
 (2) 4.5 cm downward
 (3) 1 cm downward
 (4) 2 cm downward
31. Concave mirror of focal length f (in air) is immersed in water of $\mu = \frac{4}{3}$. The focal length of the mirror will be :-
 (1) f (2) $\frac{4}{3}f$ (3) $\frac{3}{4}f$ (4) $\frac{7}{3}f$
32. The wavelength of light in two liquids 'x' and 'y' is $3500 \text{ \AA}$ and $7000 \text{ \AA}$, then the critical angle will be
 (1) 60° (2) 45°
 (3) 30° (4) 15°
33. A glass convex lens ($\mu_g = 1.5$) has a focal length of 8 cm when placed in air. What would be the focal length of the lens when it is immersed in water ($\mu_w = 1.33$)
 (1) 2 cm (2) 4 cm
 (3) 16 cm (4) 32 cm
34. If tube length of astronomical telescope is 105 cm and magnifying power is 20 for normal setting. Calculate the focal length of objective.
 (1) 100 cm (2) 10 cm
 (3) 20 cm (4) 25 cm
35. Match the corresponding entries of column-1 with column-2 (Where m is the magnification produced by the mirror) :-
- | Column-1 | Column-2 |
|---|--------------------|
| (A) $m = -2$ | (a) Convex mirror |
| (B) $m = -\frac{1}{2}$ | (b) Concave mirror |
| (C) $m = +2$ | (c) Real image |
| (D) $m = +\frac{1}{2}$ | (d) Virtual image |
| (1) $A \rightarrow b$ and c , $B \rightarrow b$ and c , $C \rightarrow b$ and d , $D \rightarrow a$ and d . | |
| (2) $A \rightarrow a$ and c , $B \rightarrow a$ and d , $C \rightarrow a$ and b , $D \rightarrow c$ and d | |
| (3) $A \rightarrow a$ and d , $B \rightarrow b$ and c , $C \rightarrow b$ and d , $D \rightarrow b$ and c | |
| (4) $A \rightarrow c$ and d , $B \rightarrow b$ and d , $C \rightarrow b$ and c , $D \rightarrow a$ and d | |
36. An astronomical telescope has a magnifying power 10. The focal length of eyepiece is 20 cm. The focal length of objective is
 (1) 2 cm (2) 200 cm
 (3) $\frac{1}{2}$ cm (4) $\frac{1}{200}$ cm
37. If a convex lens of focal length 100 cm and a concave lens of focal length 50 cm are combined together, what will be their resulting power
 (1) + 6.5 D (2) -6.5 D
 (3) + 7.5 D (4) - 1 D

SPACE FOR ROUGH WORK

- | | |
|---|--|
| <p>38. A man runs towards stationary plane mirror at a speed of 15 m/s. What is the speed of his image with respect to mirror :-
 (1) 7.5 m/s
 (2) 15 m/s
 (3) 30 m/s
 (4) 45 m/s</p> <p>39. A boy stands straight in front of a mirror at a distance of 30 cm away from it. He sees his erect image whose height is 1/5th of his real height. The mirror he is using is -
 (1) Plane mirror
 (2) Convex mirror
 (3) Concave mirror
 (4) None</p> <p>40. When a ray of light enters a medium of refractive index μ, it is observed that the angle of refraction is half of the angle of incidence is
 (1) $2 \cos^{-1}(\mu/2)$
 (2) $\cos^{-1}(\mu/2)$
 (3) $2 \cos^{-1}(\mu)$
 (4) $2 \sin^{-1}(\mu/2)$</p> <p>41. Prism angle of glass prism is 10°. It's refractive index of red and violet colour is 1.51 and 1.52 respectively. Then its dispersive power will be .
 (1) 0.015
 (2) 0.020
 (3) 0.011
 (4) 0.019</p> | <p>42. A thin Prism P_1 with angle 4° and made from glass of refractive index 1.54 is combined with another thin Prism P_2 made from glass of refractive index 1.72 to produce dispersion without deviation. The angle of Prism P_2 is
 (1) 5.33° (2) 4°
 (3) 3° (4) 2.6°</p> <p>43. Angle of prism is A and its one surface is silvered. Light rays falling at an angle of incidence $2A$ on first surface return back through the same path after suffering reflection at second silvered surface. Refractive index of the material of prism is
 (1) $2 \sin A$
 (2) $2 \cos A$
 (3) $\frac{1}{2} \cos A$
 (4) $\tan A$</p> <p>44. The radii of curvature of the two surfaces of a lens are 20 cm and 30 cm. The refractive index of lens material is 1.5. If the lens is concavoconvex then focal length of the lens is
 (1) 120 cm (2) 40 cm
 (3) 15 cm (4) 24 cm</p> <p>45. A convex lens of power P is immersed in water, how will its power change ?
 (1) Increases
 (2) Decreases
 (3) Remains unchanged
 (4) Increases for red colour and decreases for blue colour</p> |
|---|--|

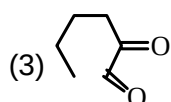
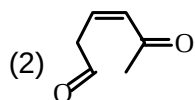
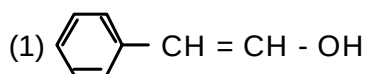
SPACE FOR ROUGH WORK

46. The 'E-isomer' is -



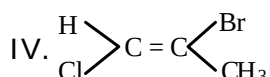
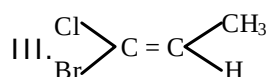
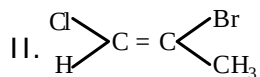
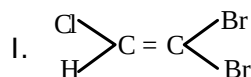
(4) none of the above

47. Tautomerism is exhibited by -



(4) All the above

48. Which is a pair of geometrical isomers ?



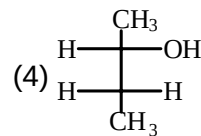
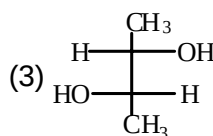
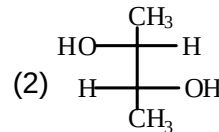
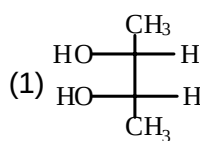
(1) I and II

(2) I and III

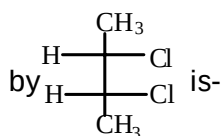
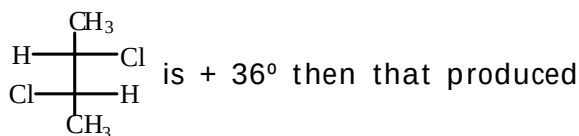
(3) II and IV

(4) III and IV

49. A Fischer projection of (2R, 3S)-2,3-butanediol is -



50. If optical rotation produced by



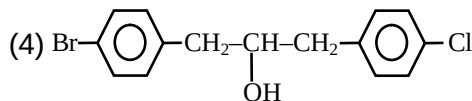
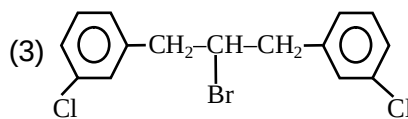
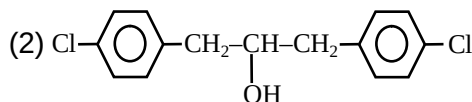
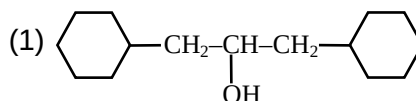
(1) -36°

(2) 0°

(3) $+36^\circ$

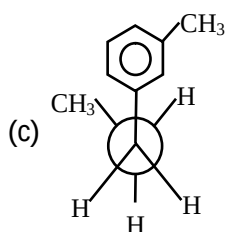
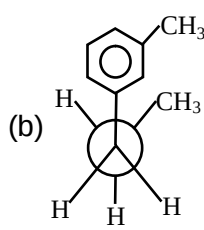
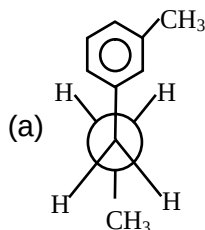
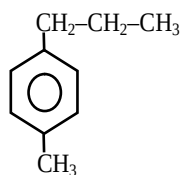
(4) Unpredictable

51. Which of the following compounds possesses an asymmetric carbon -



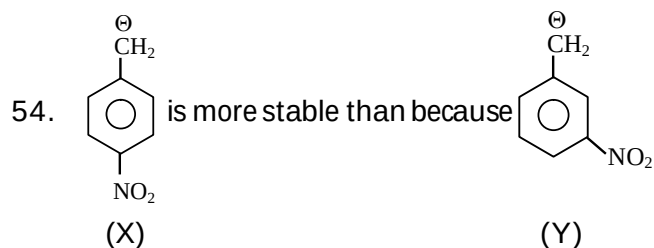
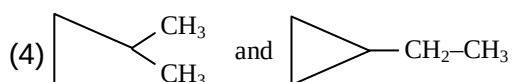
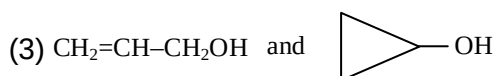
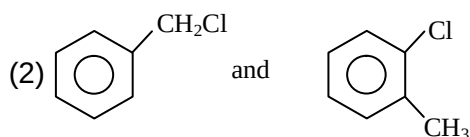
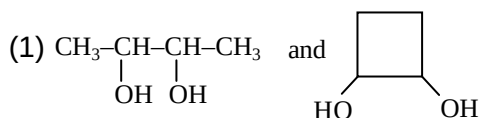
SPACE FOR ROUGH WORK

52. What is relative stability conformer of following compound ?



- (1) $a > b > c$ (2) $b > c > a$
 (3) $a = b = c$ (4) $a > c > b$

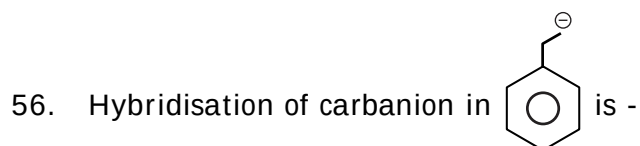
53. Which of the following pairs of compounds are ring-chain isomers?



- (1) 180° location of NO_2 and CH_2 in X
 (2) $-\text{NO}_2$ operates both $-\text{I}$ and $-\text{M}$ in X.
 (3) $-\text{I}$ in Y is weaker than X
 (4) Due to steric repulsion

55. Which of the following is most stable -

- (1) $\text{CH}_3^\ominus$ (2) $\text{CH}_2 = \text{CH}^\ominus$
 (3) $\text{CH} \equiv \text{C}^\ominus$ (4) $\text{CH}_3 - \text{C} \equiv \text{C}^\ominus$

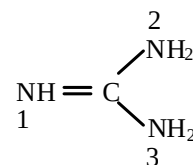


- (1) sp^3 (2) sp^2
 (3) sp (4) Unhybridised

57. Carbene is an -

- (1) Electrophile (2) Nucleophile
 (3) Very strong base (4) All may be

58. Which nitrogen is protonated readily in the guanidine ?



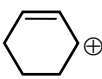
- (1) 1 (2) 2
 (3) 3 (4) None of these

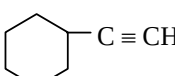
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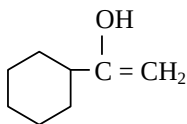
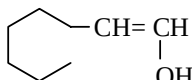
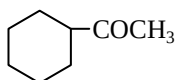
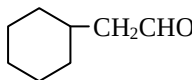
59. Arrange the following nucleophiles in the order of their nucleophilic strength :

- (1) $\text{OH}^- > \text{CH}_3\text{COO}^- > \text{OCH}_3^- > \text{C}_6\text{H}_5\text{O}^-$
 (2) $\text{CH}_3\text{COO}^- < \text{C}_6\text{H}_5\text{O}^- < \text{CH}_3\text{O}^- < \text{OH}^-$
 (3) $\text{C}_6\text{H}_5\text{O}^- < \text{CH}_3\text{COO}^- < \text{CH}_3\text{O}^- < \text{OH}^-$
 (4) $\text{CH}_3\text{COO}^- < \text{C}_6\text{H}_5\text{O}^- < \text{OH}^- < \text{CH}_3\text{O}^-$

60. Which carbocation will not form as intermediate ?

- (1) $(\text{C}_6\text{H}_5)_3\text{C}^+$ (2) 
 (3)  (4) $\text{CH}_2 = \text{C}^+\text{H}$

61. Hydration of  in presence of $\text{H}_2\text{SO}_4 / \text{HgSO}_4$ gives (as a major product)

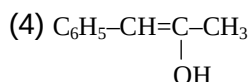
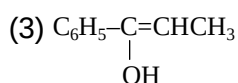
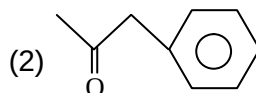
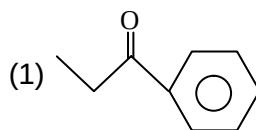
- (1)  (2) 
 (3)  (4) 

62. $\text{CH}_3\text{CH}=\text{CH}_2 \xrightarrow[\text{H}_2\text{O}_2/\text{OH}^-]{\text{BD}_3}$ product X

X is -

- (1) $\text{CH}_3 - \underset{\text{OH}}{\text{CH}} - \text{CH}_2\text{D}$
 (2) $\text{CH}_3 - \underset{\text{D}}{\text{CH}} - \text{CH}_2\text{OH}$
 (3) $\text{CH}_3 - \underset{\text{OD}}{\text{CH}} - \text{CH}_3$
 (4) None is correct

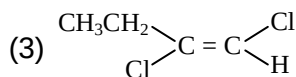
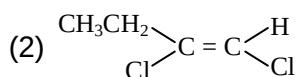
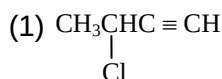
63. $\text{C}_6\text{H}_5 - \text{C} \equiv \text{C} - \text{CH}_3 \xrightarrow[\text{H}_2\text{SO}_4]{\text{HgSO}_4} \text{A}$



64. The possible products (including stereoisomers) by monochlorination of methyl cyclohexane

- (1) 4 (2) 6 (3) 8 (4) 12

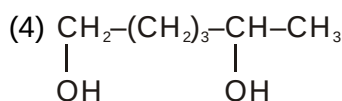
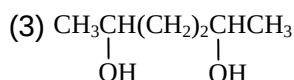
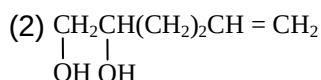
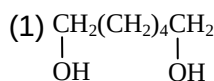
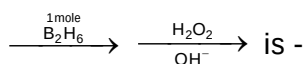
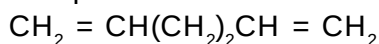
65. $\text{CH}_3\text{CH}_2\text{C} \equiv \text{CH} \xrightarrow[\text{CCl}_4]{\text{Cl}_2 (1 \text{ eq})}$ gives -



(4) None of these

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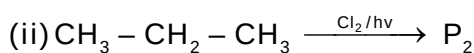
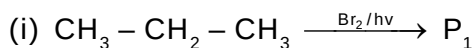
66. The product of the reaction :



67. In the chlorination of $(\text{CH}_3)_2\text{CH} - \text{CH}_2 - \text{CH}_3$ the substitution at :-

- (1) 1° carbon would be fastest
- (2) 2° carbon would be fastest
- (3) 3° carbon would be fastest
- (4) 1° , 2° , 3° carbon atoms all will occur at the same rate

68. Consider the following reaction :



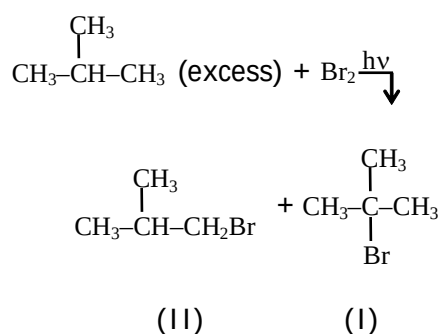
Choose the correct alternative

- (1) P_1 is 2-bromopropane while P_2 is 1-chloropropane
- (2) P_1 is 2-bromopropane and P_2 is 2-chloropropane
- (3) P_1 is 1-bromopropane mainly and P_2 is a mixture of 1-chloropropane and 2-chloropropane
- (4) P_1 is 2-Bromopropane mainly and P_2 is a mixture of 1-chloropropane and 2-chloropropane

69. The reactivity of the halogens towards halogenation of hydrocarbon decreases in the order -

- (1) $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$
- (2) $\text{I}_2 > \text{Br}_2 > \text{Cl}_2 > \text{F}_2$
- (3) $\text{F}_2 > \text{Cl}_2 > \text{I}_2 > \text{Br}_2$
- (4) $\text{Cl}_2 > \text{F}_2 > \text{Br}_2 > \text{I}_2$

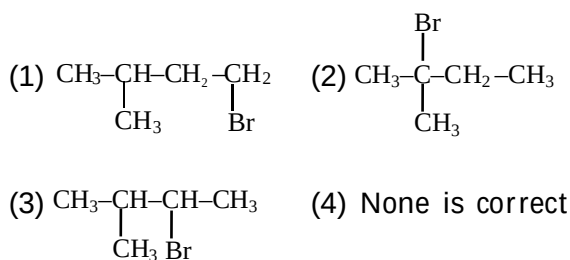
70. The relative reactivity of 1°H , 2°H and 3°H in bromination reaction has been found to be 1 : 82 : 1600 respectively. In the reaction



the percentage yields of the products (I) and (II) are expected to be -

- (1) 99.4 %, 0.6 % (2) 50%, 50%
- (3) 0.6%, 99.4% (4) 80%, 20%

71. $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3-\text{CH}-\text{CH}=\text{CH}_2 \end{array} + \text{HBr} \rightarrow \text{X (product)}$ which is predominate; X is -



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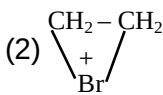
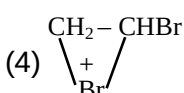
72. The reaction
 $\text{CH}_2 = \text{CH}_2 + \text{Br}_2 \longrightarrow \text{CH}_2\text{Br} - \text{CH}_2\text{Br}$
 is an example of -

- (1) Nucleophilic addition
- (2) Electrophilic addition
- (3) Elimination
- (4) Nucleophilic substitution

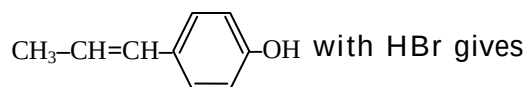
73. In the addition of HBr to propene in the absence of peroxides the first step involves the addition of -

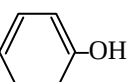
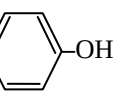
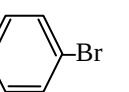
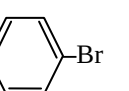
- (1) H^+ (2) Br^- (3) $\text{H}^\bullet$ (4) $\text{Br}^\bullet$

74. Ethylene reacts with Br_2 to give 1, 2-dibromoethane. The anti-addition takes place due to the formation of the intermediate -

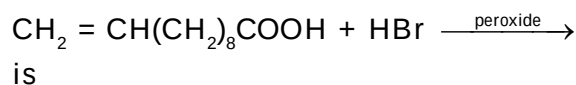
- (1) $\text{CH}_2\text{Br}-\overset{+}{\text{C}}\text{H}_2$
- (2) 
- (3) $\text{BrCH} = \text{CH}^+$
- (4) 

75. The reaction of



- (1) $\text{CH}_3-\text{CHBrCH}_2-$ 
- (2) $\text{CH}_3\text{CH}_2\text{CHBr}-$ 
- (3) $\text{CH}_3\text{CHBrCH}_2-$ 
- (4) $\text{CH}_3\text{CH}_2\text{CHBr}-$ 

76. The principal organic product formed in the reaction,



- (1) $\text{CH}_3\text{CHBr}(\text{CH}_2)_8\text{COOH}$
- (2) $\text{CH}_2 = \text{CH}(\text{CH}_2)_8\text{COBr}$
- (3) $\text{CH}_2\text{BrCH}_2(\text{CH}_2)_8\text{COOH}$
- (4) $\text{CH}_2 = \text{CH}(\text{CH}_2)_7\text{CHBrCOOH}$

77. The intermediate formed during addition of HBr to propene in presence of peroxides is :-

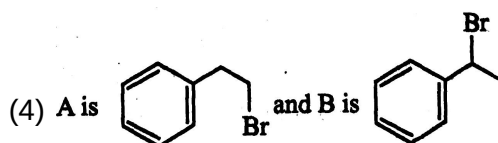
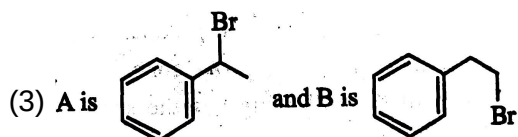
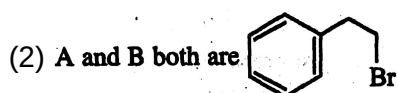
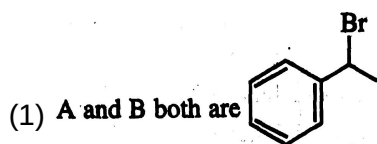
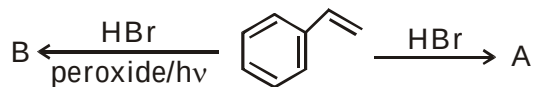
- (1) $\text{CH}_3\dot{\text{C}}\text{HCH}_2\text{Br}$
- (2) $\text{CH}_3\dot{\text{C}}^+\text{HCH}_3$
- (3) $\text{CH}_3\text{CH}_2\dot{\text{C}}^+\text{H}_2$
- (4) $\text{CH}_3\text{CHBr}\dot{\text{C}}\text{H}_2$

78. Select the reactions that would result in the formation of 2-bromopropane?

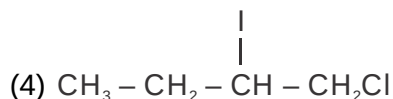
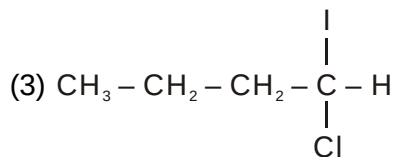
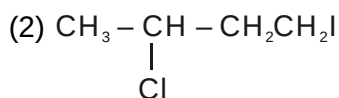
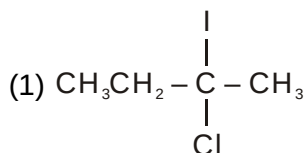
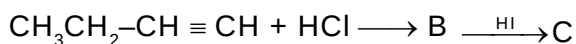
- I. $\text{CH}_3\text{CH} = \text{CH}_2 + \text{Br}_2 \xrightarrow{h\nu}$
 - II. $\text{CH}_3\text{CH} = \text{CH}_2 + \text{HBr} \xrightarrow{\text{CCl}_4}$
 - III. $\text{CH}_3\text{CH}_2\text{CH}_3 + \text{Br}_2 \xrightarrow{h\nu}$
 - IV. $\text{CH}_3\text{CH} = \text{CH}_2 + \text{Br}_2 \xrightarrow{\text{CCl}_4}$
- (1) I, II
 - (2) II, III
 - (3) I, III
 - (4) I, II, III

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79. Observe the following reactions and predict the nature of A and B :-



80. Predict the product C obtained in the following reaction of butyne-1 :-



81. In which of the following orbital diagram are both Pauli's exclusion principle and Hund's rule violated ?

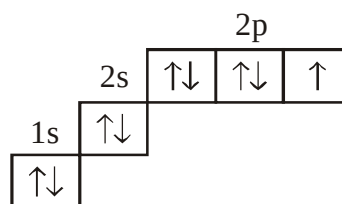


82. Radius of 2nd orbit of Li⁺² will be

(1) $0.529 \times \frac{2^2}{2} \text{ \AA}$ (2) $0.529 \times \frac{2^2}{3} \text{ \AA}$

(3) $0.529 \times \frac{3^2}{2} \text{ \AA}$ (4) $0.529 \times \frac{3^2}{3} \text{ \AA}$

83. Which element is represented by the following electronic configuration



(1) Nitrogen

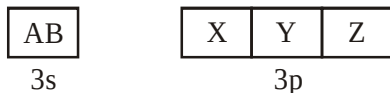
(2) Oxygen

(3) Fluorine

(4) Neon

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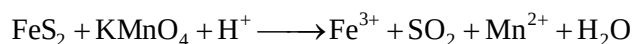
84. Five valence electrons of $_{15}\text{P}$ are labelled as



If the spin quantum of B and Z is $+\frac{1}{2}$, the group of electrons with three of the quantum number same are

- (1) AB, XYZ, BY
(2) AB, AY
(3) XYZ, AZ
(4) AB, XYZ, AY
85. Oxidation number of Cl in CaOCl_2 is
(1) -1 and $+1$
(2) $+2$
(3) -2
(4) None of these
86. The oxidation number of sulphur in S_8 , S_2F_2 and H_2S respectively are
(1) 0 , $+1$ and -2
(2) $+2$, $+1$ and -2
(3) 0 , $+1$ and $+2$
(4) -2 , $+1$ and -2
87. The oxide which cannot act as a reducing agent, is
(1) SO_2
(2) NO_2
(3) CO_2
(4) ClO_2

88. In the reaction,



the equivalent mass of FeS_2 would be equal to

- (1) Molar mass
(2) $\frac{\text{Molar mass}}{10}$
(3) $\frac{\text{Molar mass}}{11}$
(4) $\frac{\text{Molar mass}}{13}$
89. Consider the following reaction :
- $$\text{P}_4 + 3\text{NaOH} + 3\text{H}_2\text{O} \longrightarrow 3\text{NaH}_2\text{PO}_2 + \text{PH}_3$$
- In the above reaction,
1. Phosphorus is oxidised.
2. Phosphorus is reduced.
Which of the above is/are correct ?
(1) 1 only
(2) 2 only
(3) Both 1 and 2
(4) Neither 1 nor 2
90. Chromium has the electronic configuration $4s^1 3d^5$ rather than $4s^2 3d^4$ because
(1) $4s$ and $3d$ have the same energy
(2) $4s$ has a higher energy than $3d$
(3) $4s^1$ is more stable than $4s^2$
(4) half-filled subshell is more stable

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91. Which of the following is true for proteins which remain partially or totally buried in the plasma membrane ?

- (1) All are involved in active transport of charged & polar molecules across the membrane
- (2) These cannot be easily extracted
- (3) These can be easily extracted
- (4) These represent only 30% of the total membrane protein

92. Match the column :

Column-I		Column-II	
A.	Sap-vacuole	(i)	Buoyancy
B.	Contractile-vacuole	(ii)	Anthocyanin
C.	Food-vacuole	(iii)	Excretion
D.	Gas-vacuole	(iv)	Engulfing of food

- (1) A = (i), B = (iv), C = (iii), D = (ii)
- (2) A = (ii), B = (iii), C = (iv), D = (i)
- (3) A = (ii), B = (iii), C = (i), D = (iv)
- (4) A = (iii), B = (i), C = (iv), D = (ii)

93. According to the unit membrane model, the thickness of cell membrane is :

- (1) 75 Å
- (2) 35 Å
- (3) 20 Å
- (4) 25 Å

94. Read the following statements :

- (i) Peroxisomes are involved in photorespiration of the non-green cell of plants and also perform β -oxidation of fatty acids in animal cells
- (ii) Mitochondria possess enzymes for the β -oxidation of fatty acids
- (iii) The fluid-dynamic mosaic model of plasma membrane explains both structural and functional aspects
- (iv) The movement of cilia and flagella is due to the presence of dynein proteins
- (v) Lysosome are formed by process of packaging in the golgi apparatus.

How many of the above statements are correct?

- (1) Two
- (2) Three
- (3) Four
- (4) Five

95. Which of the following is not incorrect ?

- (1) The ribosome of chloroplast is larger than the cytoplasmic ribosome
- (2) The inner membrane of chloroplast is relatively less permeable in comparison to the outer membrane

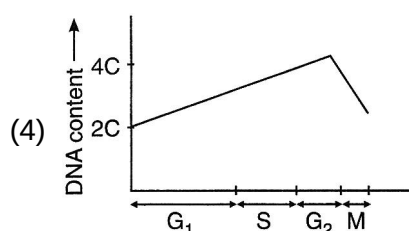
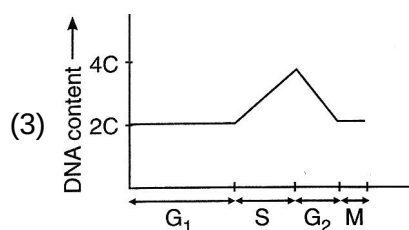
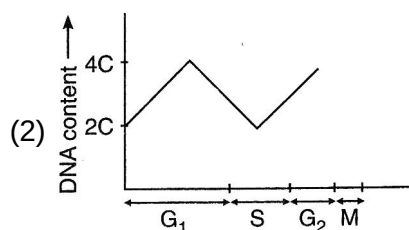
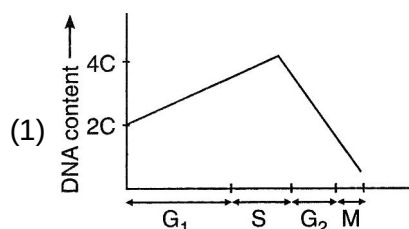
(3) The space limited by the inner membrane of chloroplast is called intermembrane space

(4) The lumen of thylakoid contain enzymes

96. If a mitocyte has 10 pg of DNA and 16 chromosomes in G_1 then which of the following is correct ?

- (1) S-phase will have 5 pg of DNA
- (2) The cell at anaphase stage will have 20 pg of DNA and a total of 32 chromosomes
- (3) G_2 -phase will have 8 chromosomes in a cell
- (4) The cell at metaphase will have 10 pg of DNA and a total of 16 chromosomes

97. Which of the following graphs indicate the relative change in the amount of chloroplast DNA of a cell showing mitosis ?



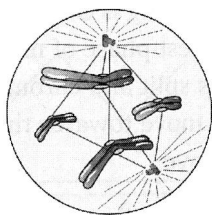
98. Phragmoplast :

- (1) Helps in cell enlargement
- (2) Formation occur during mitotic anaphase
- (3) Formation occur during cytokinesis of bacterial cell
- (4) Help in plant cytokinesis

99. Identify the wrong statement about meiosis:

- (1) Pairing of homologous chromosomes
- (2) Four haploid cells are formed
- (3) At the end of meiosis number of chromosomes are reduced to half
- (4) Two cycles of DNA replication occur

100. Choose the correct option for the diagram given below:



	Number of Chromosomes	Number of Chromatids	Number of Centromeres	Number of Kinetochores
(1)	4	8	8	8
(2)	4	4	4	8
(3)	4	16	8	4
(4)	4	8	4	8

101. Skin colour in human, is controlled by three gene pairs, then what is the number of intermediate & parental combinations in F_2 generation respectively :-

- (1) 2, 62
- (2) 62, 2
- (3) 2, 14
- (4) 14, 2

102. Which of the following type of inheritance can we find out with the help of reciprocal cross?

- (1) Cytoplasmic inheritance
- (2) Gene inheritance
- (3) Maternal inheritance
- (4) (1) & (3) both

103. Genes a, b and c assort independently and are recessive to their respective alleles A, B and C. Two triple heterozygous AaBbCc individual are crossed what is the probability that a given offspring will be genotypically homozygous for all three dominant alleles?

- (1) $2/64$
- (2) $27/64$
- (3) $1/64$
- (4) $6/64$

104. A dihybrid plant is with genotype GgTt. In this plant first gene pair is codominant on selfing of this plant what is phenotypic ratio?

- (1) 2 : 1
- (2) 3 : 1
- (3) 3 : 6 : 3 : 1 : 2 : 1
- (4) 6 : 3 : 2 : 1

105. The allele, which is unable to express its effect in the presence of another is called :-

- (1) Co-dominant
- (2) Supplementary
- (3) Complementary
- (4) Recessive

106. Based on observations on monohybrid crosses Mendel draw some conclusion. Which of the following is not correct?

- (1) Characters are controlled by discrete units called factors
- (2) Factors occur in pairs
- (3) In a similar pair of factors one member of the pair dominates the other
- (4) The postulate of dominance also explains the proportion of 3 : 1 obtained at the F_2

107. In the inheritance of flower colour in dog flower plant the F_1 had a phenotype that

- (1) Resembles both of the parents
- (2) Did not resemble either of the two parents
- (3) Resemble with only one parent
- (4) (1) & (3) both

108. The three different alleles of human ABO blood types will produce how many genotypes & phenotypes respectively :-
 (1) 4 & 6 (2) 6 & 4
 (3) 6 & 6 (4) 4 & 4
109. Other than pea plants it was found that sometimes the F_1 had a phenotype that did not resemble either of the two parents and was in between the two. Due to :-
 (1) Complete dominance
 (2) Incomplete dominance
 (3) Co-dominance
 (4) Complementary gene interaction
110. Gene which code for a pair of contrasting traits are known as :-
 (1) Allele
 (2) Non allele
 (3) Pseudo allele
 (4) Isoallele
111. If dominant C and P genes are essential for the development of purple colour in *Lathyrus odoratus* flowers. What would be the ratio of purple and white colour in a cross between $CcPp \times ccPp$.
 (1) 3 : 5 (2) 9 : 7
 (3) 2 : 6 (4) 4 : 4
112. Which of the following condition is true for codominance :-
 (1) Phenotype of F_1 resembled either of the two parents
 (2) Phenotype of F_1 did not resemble either of two parents
 (3) Phenotype of F_1 resembles both parents
 (4) None of these
113. Which of the following is a good example of multiple allele :-
 (1) ABO blood group
 (2) Size of starch grain in pea
 (3) Shape of seed
 (4) Flower colour in pea

114. A man having the genotype $AABbCcDD$ can produce P number of genetically different sperms, and a woman of genotype $LLMmNn$ can generate Q number of genetically different eggs. Determine the values of P and Q :-
 (1) $P = 4, Q = 4$
 (2) $P = 4, Q = 8$
 (3) $P = 8, Q = 4$
 (4) $P = 8, Q = 8$
115. In mice, black coat colour (allele B) is dominant to brown coat colour (allele b). The offspring of a cross between a black mouse (BB) and a brown mouse (bb) were allowed to interbreed. What percentage of the progeny would have black coats in F_2 generation?
 (1) 25% (2) 50%
 (3) 75 % (4) 100 %
116. Fruit shape in shepherd's purse (*Capsella bursa*) is of two types- triangular and top-shaped. Triangular fruit shape (T) is dominant over top-shape (t). Following table summarizes the results of several crosses.

Cross	Result
Strain 1 $\times$ tt	All triangular
Strain 2 $\times$ tt	1 triangular : 1 top-shaped
Strain 3 $\times$ tt	All top-shaped
Strain 4 $\times$ Tt	3 triangular : 1 top-shaped

Which pair of strains possess the genotype Tt?

- (1) Strains 2 and 3
 (2) Strains 2 and 4
 (3) Strains 1 and 3
 (4) Strains 1 and 4
117. Which Mendelian cross can produce two genotypes and two phenotypes
 (1) Monohybrid cross
 (2) Monohybrid test cross
 (3) Incomplete dominance
 (4) Codominance

118. Which one of the following crosses would have 1 : 1 : 1 : 1 ratio ?

- (1) $TtRR \times ttrr$
- (2) $TTRR \times ttrr$
- (3) $TtRr \times ttrr$
- (4) $TtRR \times TTrr$

119. Possible blood group in children from the parents with 'B' and 'O' blood group are :

- (1) All B
- (2) All O
- (3) Both B and O
- (4) A and B

120. Complementary genes were demonstrated by Bateson and Punnett in

- (1) *Capsella*
- (2) *Lathyrus odoratus*
- (3) *Antirrhinum*
- (4) *Mirabilis*

121. Which of the following genotype of sweet pea plant is related with the production of purple coloured flowers ?

- (1) $CcPp$ (2) $CCpp$
- (3) $ccPP$ (4) $Ccpp$

122. The genotypes of a husband and Wife are $I^A I^B$ & $I^A i$. Among the blood types of their children, how many different genotypes and phenotypes are possible?

- (1) 3 genotypes ; 4 phenotypes
- (2) 4 genotypes ; 3 phenotypes
- (3) 4 genotypes ; 4 phenotypes
- (4) 3 genotypes ; 3 phenotypes

123. A true breeding plant is :

- (1) homozygous and produces offspring of its own kind
- (2) always homozygous recessive in its genetic constitution
- (3) one that is able to breed on its own
- (4) produced due to cross-pollination among unrelated plants

124. Match the terms in Column-I with their description in Column-II and choose the correct option :

Column-I		Column-II	
(a)	Dominance	(i)	Many genes govern a single character
(b)	Codominance	(ii)	In a heterozygous organism only one allele expresses itself
(c)	Pleiotropy	(iii)	In a heterozygous organism both alleles express themselves fully
(d)	Polygenic inheritance	(iv)	A single gene influences many characters

- | | | | |
|----------|-------|------|-------|
| (a) | (b) | (c) | (d) |
| (1) (ii) | (i) | (iv) | (iii) |
| (2) (ii) | (iii) | (iv) | (i) |
| (3) (iv) | (i) | (ii) | (iii) |
| (4) (iv) | (iii) | (i) | (ii) |

125. If two persons with 'AB' blood group marry and have sufficiently large number of children, these children could be classified as 'A' blood group : 'AB' blood group 'B' blood group in 1 : 2 : 1 ratio. Modern technique of protein electrophoresis reveals presence of both 'A' and 'B' type proteins in 'AB' blood group individuals. This is an example of :

- (1) Complete dominance
- (2) Codominance
- (3) Incomplete dominance
- (4) Partial dominance

126. Which Mendelism idea is depicted by a cross in which the F_1 generation resembles both the parents ?

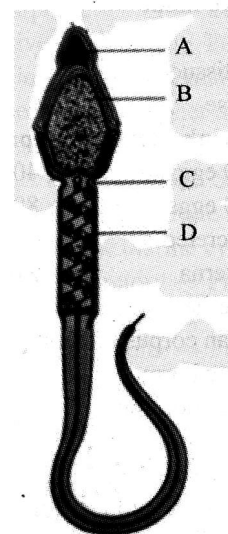
- (1) co-dominance
- (2) incomplete dominance
- (3) law of dominance
- (4) inheritance of one gene

127. A certain road accident patient with unknown blood group needs immediate blood transfusion. His one doctor friend at once offers his blood. What was the blood group of the donor ?
- (1) Blood group O
 - (2) Blood group A
 - (3) Blood group B
 - (4) Blood group AB
128. In *Antirrhinum* two plants with pink flowers were hybridized. The F₂ plants produced red, pink and white flowers in the proportion of 1 red, 2 pink and 1 white. What could be the genotype of the two plants used for hybridization ? Red flower colour is determined by RR, and white by rr genes :-
- (1) RR
 - (2) Rr
 - (3) rr
 - (4) rrrr
129. A cross in which an organism showing a dominant phenotype is crossed with the recessive parent in order to know its genotype is called :-
- (1) Back cross
 - (2) Test cross
 - (3) Dihybrid cross
 - (4) Monohybrid cross
130. Sickle cell anemia is :-
- (1) Characterized by elongated sickle like RBCs with a nucleus
 - (2) An autosomal linked dominant trait
 - (3) Caused by substitution of valine by glutamic acid in the beta globin chain of haemoglobin
 - (4) Caused by a change in a single base pair of DNA
131. The most popularly known blood grouping is the ABO grouping. It is named ABO and not ABC, because "O" in it refers to having :-
- (1) No antigens A and B on RBCs
 - (2) Other antigens besides A and B on RBCs
 - (3) Overdominance of this type on the genes for A and B types
 - (4) One antibody only - either anti-A or anti-B on the RBCs
132. Inheritance of skin colour in humans is an example of :-
- (1) Codominance
 - (2) Chromosomal aberration
 - (3) Point mutation
 - (4) Polygenic inheritance
133. In pea plants, yellow seeds are dominant to green. If a heterozygous yellow seeded plant is crossed with a green seeded plant, what ratio of yellow and green seeded plants would you expect in F₁ generation?
- (1) 3 : 1
 - (2) 50 : 50
 - (3) 9 : 1
 - (4) 1 : 3
134. Genotype of father is I^AI^BRr & mother is I^AI^ORr what will be the probability of their child have AB⁺ blood group.
- (1) $\frac{3}{16}$
 - (2) $\frac{4}{15}$
 - (3) $\frac{5}{18}$
 - (4) $\frac{5}{16}$
135. A roan bull is crossed with 3 cow A, B, C in which cow A black, cow B, roan, cow C white coloured.
- In all three crosses the roan offspring obtained in the proportion of :-
- (1) 0 : 1 : 2
 - (2) 1 : 0 : 1
 - (3) 1 : 2 : 1
 - (4) 2 : 3 : 1

- | | |
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| <p>136. Birth canal comprises of
 (1) Vaginal canal only
 (2) Cervical canal only
 (3) Fallopian tube
 (4) Vaginal canal and cervical canal</p> <p>137. The space between two folds of labia minora is called
 (1) Perineum
 (2) Vestibule
 (3) Clitoris
 (4) Foreskin</p> <p>138. In females, an erectile body homologous with penis of male is
 (1) Prepuce (2) Fourchette
 (3) Clitoris (3) Uterus</p> <p>139. Which of the following is not a male secondary sexual character?
 (1) Enlarged penis
 (2) Hoarse voice
 (3) Beard and moustache
 (4) Deposition of fats in buttocks</p> <p>140. Mammary glands are
 (1) Modified sebaceous gland
 (2) Modified sweat gland
 (3) Homologous to male prostate
 (4) Endocrinal glands</p> <p>141. Deposition of fats in the breast, hips as well as growth of pubic hair during puberty is under the influence of
 (1) Androgen
 (2) Progesterone
 (3) Estrogen
 (4) LH</p> <p>142. The vaginal opening remains partially covered by a membranous
 (1) Fimbriae
 (2) Hymen
 (3) Perineum
 (4) Vestibule</p> <p>143. Among clitoris, hymen, vagina, cervix, labia majora, labia minora and mons pubis, how many are part of external genitalia (Vulva) in females?
 (1) 3 (2) 5 (3) 2 (4) 6</p> | <p>144. Most confirmatory forensic test for rape is
 (1) Presence of glucose in vagina
 (2) Presence of hymen
 (3) Presence of fructose in vagina
 (4) Presence of LH in urine</p> <p>145. Which of the following cells undergoes second meiotic division?
 (1) Spermatogonia
 (2) Spermatids
 (3) Primary spermatocyte
 (4) Secondary spermatocyte</p> <p>146. Surgical removal of prepuce/foreskin is called
 (1) Castration
 (2) Circumcision
 (3) Cryptorchidism
 (4) Orchidectomy</p> <p>147. Energy for sperm motility is provided by
 (1) Head
 (2) Tail
 (3) Neck
 (4) Middle piece</p> <p>148. The release of sperms from seminiferous tubule is called
 (1) Semination
 (2) Spermiation
 (2) Ejaculation
 (4) Insemination</p> <p>149. Acrosomal cap of sperm is derived from
 (1) Mitochondria
 (2) Centriole
 (3) Golgi bodies
 (4) Nucleus</p> <p>150. All of the following are present in acrosome of sperm, except
 (1) Acrosin
 (2) Zona lysine
 (3) Corona penetrating enzyme
 (4) Hyaluronic acid</p> |
|---|---|

151. Which of the following undergoes the first meiotic division leading to formation of two equal, haploid cells with 23 chromosomes?
 (1) Spermatogonia
 (2) Secondary spermatocyte
 (3) Primary spermatocyte
 (4) Spermatid
152. The number of sperms produced from 24 secondary spermatocytes will be
 (1) 24 (2) 12 (3) 48 (4) 96
153. Seminal plasma in human males is rich in
 (1) Glucose and calcium
 (2) Fructose and calcium
 (3) Ribose and sodium
 (4) DNA and sodium
154. How many sperms are ejaculated by human male during a coitus?
 (1) 200 - 300 billion (2) 20-30 million
 (3) 200 - 300 million (4) 500 million
155. The distal centriole gives attachment to the axial filament but the proximal helps in
 (1) Motility
 (2) Cleavage of fertilized egg
 (3) Provides energy
 (4) Helps distal centriole
156. Tubuli recti connect
 (1) Rete testis and epididymis
 (2) Two seminiferous tubules
 (3) Epididymis and vas deferens
 (4) Seminiferous tubule and rete testis
157. Spermiogenesis is facilitated by secretion of some factors from _____.
 (1) Leydig cells (2) Sertoli cells
 (3) Interstitial cells (4) Spermatogia
158. Seminal plasma is not contributed by
 (1) Prostate gland
 (2) Seminal vesicle
 (3) Bulbourethral gland
 (4) Bartholin's gland
159. Reduced motility of sperms is called _____
 (1) Azoospermia
 (2) Oligospermia
 (3) Asthenozoospermia
 (4) Teratozoospermia

160. Which of the following is incorrect regarding semen?
 (1) 200 - 300 million sperms are released during ejaculation.
 (2) Milky appearance of semen is due to secretions from prostate gland.
 (3) Semen is slightly acidic with pH 6.7.
 (4) The seminal plasma along with sperms constitute semen.
161. Choose the correct answer which shows all accurate labelling of human sperm.



- (1) A - Acrosome , B - Neck, C - Axial filament, D - Nucleus
 (2) A - Acrosome, B - Proximal centriole, C - Mitochondria, D - Nucleus
 (3) A-Acrosome, B-Head, C-Neck, D - Mitochondria
 (4) A - Acrosome, B - Proximal centriole, C -Axial filament, D - Nucleus
162. The male accessory ducts include
 (1) Rete testis, vasa efferentia, epididymis and vas deferens
 (2) Vasa efferentia, ejaculatory duct, urethra
 (3) Seminiferous tubules, vasa efferentia, epididymis, vas deferens
 (4) Epididymis and vasa efferentia only

163. Seminiferous tubules are lined by two types of cells, i.e.
- (1) Sertoli cells and Leydig cells
 - (2) Sertoli cells and Nurse cells
 - (3) Spermatogonial cells and Sertoli cells
 - (4) Spermatogonial cells and Leydig cells
164. The hormone responsible for the descend of testis into scrotal sac is
- (1) LH
 - (2) Testosterone
 - (3) ICSH
 - (4) Progesterone
165. In testis, the interstitial spaces are regions outside the seminiferous tubules and contain
- (1) Leydig cells
 - (2) Sertoli cells
 - (3) Immunologically competent cells
 - (4) Both (1) and (3)
166. The part of fallopian tube closer to ovary is
- (1) Isthmus
 - (2) Cervix
 - (3) Ampulla
 - (4) Infundibulum
167. If vas deferens of both sides are cut, then
- (1) Gamete production is inhibited
 - (2) There is no desire for sexual intercourse
 - (3) Sperm count remains same
 - (4) Semen is without sperms
168. Which of the following is the correct path of movement of sperms?
- (1) Seminiferous tubule → vasa efferentia → rete testis → caput epididymis → tubuli recti
 - (2) Seminiferous tubule → rete testis → vasa efferentia → tubuli recti → cauda epididymis
 - (3) Seminiferous tubule → tubuli recti → rete testis → vasa efferentia → caput epididymis → corpus epididymis
 - (4) Seminiferous tubuli → caput epididymis → vasa efferentia → corpus epididymis → rete testis
169. Mammals having intra abdominal testes are
- (1) Marsupials and whale
 - (2) Cat, dog and whale
 - (3) Egg laying mammals, whale and elephant
 - (4) Marsupials and elephant
170. Which of the following function of Sertoli cell mediates a negative feedback control of Follicle Stimulating Hormone (FSH) secretions?
- (1) Formation of testosterone
 - (2) Formation of anti-mullerian factor
 - (3) Formation of blood testis barrier
 - (4) Synthesis of inhibin
171. A human female starts developing male characteristic like beard, degeneration of uterus and ovaries, enlargement of clitoris etc. It may be due to :-
- (1) Damage of posterior pituitary
 - (2) Over production of adrenal androgens
 - (3) Surgical removal of mammary glands
 - (4) Over production of aldosterone and testosterone
172. Match list-I with II and select the correct option:-
- | List-I | | List-II | | |
|-------------------------|-----|------------------------|-----|-----|
| (a) Adrenalin | | (1) Myxoedema | | |
| (b) Hyperparathyroidism | | (2) heart beat | | |
| (c) Oxytocin | | (3) Salt-water balance | | |
| (d) Hypothyroidism | | (4) Child birth | | |
| (e) Aldosterone | | (5) Demineralization | | |
| (1) a-1 | b-5 | c-4 | d-2 | e-3 |
| (2) a-2 | b-5 | c-4 | d-1 | e-3 |
| (3) a-2 | b-3 | c-4 | d-1 | e-5 |
| (4) a-4 | b-3 | c-2 | d-1 | e-5 |

173. In hormone action, if receptor molecules are removed from target organ, the target organ will-
- (1) Continue to respond to hormone
 - (2) Not respond to hormone
 - (3) continue to respond but require higher Concentration
 - (4) Continue to respond but in the opposite way.
174. Which of these hormones uses cAMP as a second messenger ?
- (1) Estrogen
 - (2) Cortisol
 - (3) Epinephrine
 - (4) Testosterone
175. In a hormonal control system, hormone X, promote the secretion of substance Y. If this system works by negative feedback, which of the following be correct ?
- (1) Substance Y stimulates the production of hormone X
 - (2) Secretion of Y substance should stop completely.
 - (3) Level of hormone X should not be affected by level of substance Y.
 - (4) If level of substance Y highly increases secretion of hormone X should decrease
176. The function of insulin hormone is :-
- (1) To release glucose from liver cells and glycogenolysis promotion
 - (2) To increase cellular glucose uptake and consumption
 - (3) To increase blood sugar level
 - (4) To increase lipolysis
177. Oxytocin responsible for vigorous contraction of uterus at the time of child birth is synthesize by:-
- (1) posterior pituitary gland
 - (2) Hypothalamus
 - (3) Both 1 and 2
 - (4) Anterior pituitary gland
178. Hormones involved in carbohydrate metabolism are :-
- (1) Insulin, glucagon, epinephrine, PTH
 - (2) Insulin, glucagon, epinephrine, glucocorticoids
 - (3) Insulin, glucagon, calcitonin, glucocorticoids
 - (4) Insulin, glucagon, epinephrine, Melatonin
179. Which of the following statements are correct ?
- (a) Cortisol suppresses the immune response
 - (b) Thyroxine supports the process of red blood cell formation
 - (c) Glucagon stimulates glycogenesis process
 - (d) Androgens stimulate muscular growth, aggressiveness, high pitch of voice.
 - (e) Insulin decreases breakdown of glycogen
- (1) a, b, c are true, but d, e are false
 - (2) a, b, d are true, but c, e are false
 - (3) d, e are true, but a, b, c are false
 - (4) a, b, e are true, but c, d are false
180. Match the function of hormone in column I with their hormone in column II
- | Column I | Column II |
|--|------------------|
| (A) Produces anti inflammatory reactions | (i) Prolactin |
| (B) Regulation of Basal Metabolic rate | (ii) Aldosterone |
| (C) Stimulates reabsorption of Na ⁺ | (iii) Insulin |
| (D) Stimulates glycogenesis process | (iv) ADH |
| (E) Regulates growth of the mammary gland | (v) Cortisol |
| (F) Stimulates reabsorption of water | (vi) Thyroxine |
- (1) A-(v), B-(vi), C-(ii), D-(iii), E-(i), F-(iv)
 - (2) A-(v), B-(vi), C-(iv), D-(iii), E-(i), F-(ii)
 - (3) A-(vi), B-(iii), C-(ii), D-(v), E-(i), F-(iv)
 - (4) A-(vi), B-(iii), C-(i), D-(iv), E-(ii), F-(v)